

CODE SPORT



In this month's column, we discuss a few more questions on machine learning and deep learning.

As we have been doing over the last couple of months, we will continue to discuss a few more interview questions in this month's column as well, focusing on topics in machine learning. Let us start off with a simple question.

1. Let us assume that you have a neural network in which there is only an input layer (with a single input X being a number) and an output layer with a single output neuron. There are no hidden layers. You can assume any kind of output layer activation, be it linear, sigmoid, softmax, tanh or ReLU . The output needs to be 0 for $X < 6$ and 1 for $X > 6$. How would you design your neural network?
2. Is it possible for a classifier to have both high bias and high variance? In that case, would the training set error be greater than the test set error or vice versa? How can you recognise such a situation—whether it is possible for a classifier to have both high bias and high variance?
3. You are working on developing a part of a large image classification system of indoor household images taken from different cameras placed inside the house. Such a system can be part of a smart home project or for home security considerations. Since many of your customers have pet cats, your system needs to be able to recognise cats. You have designed a machine learning system that can recognise cat images from non-cat images. Note that the household cameras being used in the setup are quite cheap, and can end up producing poor quality images due to variations in lighting, camera motion, etc. Since there are zillions of cat images on the Internet, you have decided to train your system using pet images from the Internet, which are typically high quality. You have downloaded 100,000 labelled pet images from the Internet to train your cat-recognition classifier. You have also obtained 5000 home-camera generated

images, which may include cat images as well. You are planning to use 2500 of these 5000 images as the validation set for tuning the hyper-parameters. The remaining 2500 images would be used as the test set for final evaluation. After you have finished developing the initial model and are about to start the next step, which is the hyper-parameter tuning exercise, your friends tell you that your methodology is incorrect because your training set and validation set come from different data distributions and, hence, it is likely that your chosen model would perform badly on the test set. Would you agree with them? If not, explain why. If yes, explain how you would change your methodology. Note that you cannot increase the amount of indoor household camera image data, which is only 5000 images, of which a few hundred are cat images.

4. You are designing a classifier which will analyse data relating to Bengaluru traffic fines and build a classification system which will predict whether the fine for a particular violation will be paid or not. The input data set has information on the offender, such as the name, residential street name, house number, locality pin code, whether the person is a repeat offender, age, gender, information regarding what type of traffic violation occurred, the date of violation, where it happened, information regarding the fine such as the amount, any late fee, administrative charges, etc. Note that the different variables in the input data set are of different data types, with the type of traffic violation represented in string format, the residential street number being a numerical quantity, and the amount of the fine being a floating point number. The target variable you are predicting is a binary variable which is 1 if the fine will be paid, or 0 if the fine will not be paid. You have carefully analysed the data